

Two transverse waveforms (A and B) travel through the same medium with a phase difference of 180° . If the amplitude of the waveforms is ± 100 m, what is the combined displacement of both waveforms as waveform A reaches maximum displacement?

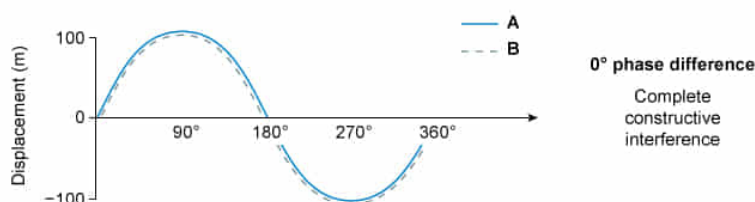
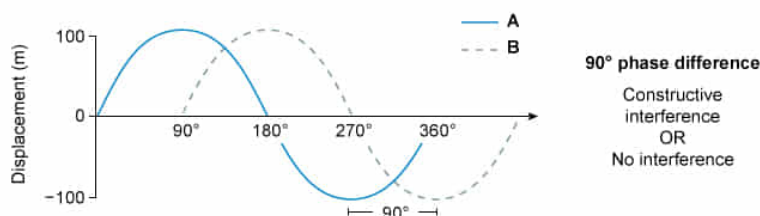
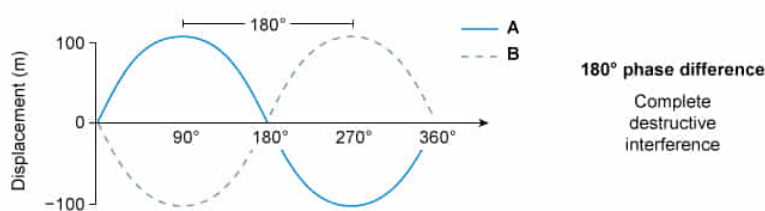
- ✓ ☒ A. 0 m [69%]
☐ B. 100 m [6%]
☐ C. 175 m [3%]
☐ D. 200 m [19%]

Correct

69% answered correctly

Explanation:

Effect of waveform phase difference on interference



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Mechanical waves transmit energy by displacing matter

through temporary, periodic oscillations (ie, vibrations).

Transverse mechanical waves are modeled by sinusoidal functions that express the momentary displacement of the propagation medium (y-axis) through a cyclical interval (x-axis).

Sinusoidal functions are used to model mechanical waves because each function contains terms analogous to the fundamental **properties** of waves (eg, frequency, amplitude, periodicity). In particular, sinusoidal functions used to model mechanical waves repeat every 360° (or 2π radians), which is one complete cycle. For example, a wave function yielding a maximum displacement at 90° will **once again** yield maximum displacement at 450° following another complete cycle ($90^\circ + 360^\circ = 450^\circ$).

Phase difference (or phase shift) refers to the degree to which the cyclical nature of two equal-frequency waveforms are aligned. For example, a phase difference of 0° indicates that both wave functions are fully overlaid and reach minimum or maximum displacement together at the same point in the waveform cycle. Conversely, a phase difference of 180° (half a cycle) indicates that the maximum of one wave function occurs concurrent to the minimum of the other waveform. As a result, phase difference expresses the degree of constructive and destructive **interference** between two waveforms.

In this question, waveforms A and B are out of phase by 180° . Therefore, when waveform A reaches its *maximum* displacement (+100 m) waveform B reaches its *minimum* displacement (-100 m).

The expression for total displacement caused by two interfering waveforms is the sum of the individual relative displacements (y-axis values) of each waveform:

$$\text{Total displacement} = \text{Waveform A displacement} + \text{Waveform B displacement}$$

$$\text{Total displacement} = (100 \text{ m} + -100 \text{ m}) = 0 \text{ m}$$

Therefore, the combined displacement of both waveforms as waveform A reaches maximum is 0 m.

Educational objective:

Phase difference relates the alignment of two equal-frequency waveforms to one another. Phase differences can result in constructive interference, destructive interference, or no interference.

Time spent:0 sec

Subject:
Physics

QID:402289

System:
Light and Sound

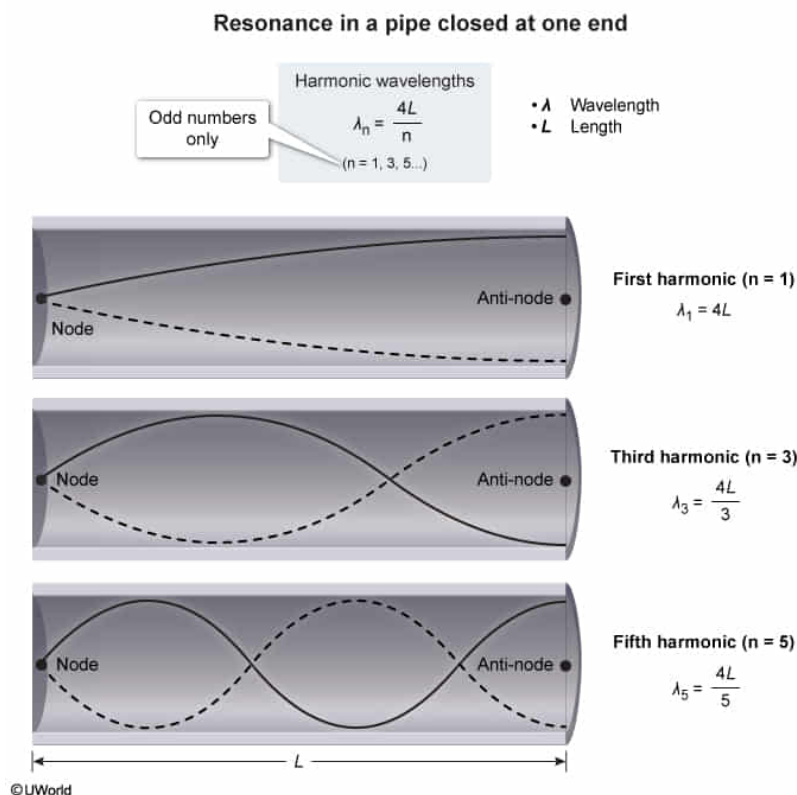
What expression gives the wavelength of the third harmonic for a 1.5 m pipe that is closed at one end?

- ☐ A. $(2) \times (1.5) / 3$ [19%]
- ✓ ☒ B. $(4) \times (1.5) / 3$ [58%]
- ☐ C. $(2) \times (1.5) / 5$ [4%]
- ☐ D. $(4) \times (1.5) / 5$ [17%]

Correct

59% answered correctly

Explanation:



The shape and dimensions of a pipe determine the properties of **standing waves** created within it. Typically, standing waves with various wavelengths, known as harmonics, are simultaneously generated within a pipe. The unique acoustic qualities of a wind instrument (ie, a musical instrument built

from an open or closed pipe) are derived from the unique harmonic series of that particular instrument.

The **wavelength of the first harmonic (λ_1)** corresponds to the soundwave of the longest wavelength that may be produced within a pipe. Furthermore, **higher harmonics** have smaller wavelengths and higher frequencies than those of the first harmonic.

For a pipe closed at one end, the wavelength of each harmonic is proportional to **object length (L)** and **harmonic number ($n = 1, 3, 5\dots$)**:

$$\lambda_n = \frac{4L}{n}$$

Pipes closed at one end cannot produce **even-numbered harmonics** ($n = 2, 4, 6\dots$), as no displacement of air takes place at the closed end. In other words, the closed end of the pipe is always a **node** (ie, a point of minimum displacement) and the open end of the pipe is always an **antinode** (ie, a point of maximum displacement).

Nevertheless, even-numbered harmonics retain a place within the harmonic series. Consequently, the harmonic number associated with the second-longest standing wave of a pipe closed at one end is not 2 but 3. In such a case, the expression that gives the wavelength of the third harmonic is:

$$\lambda_3 = \frac{4L}{3}$$

Therefore, the expression that gives the wavelength of the

third harmonic for a 1.5 m pipe that is closed at one end is:

$$\lambda_3 = \frac{4(1.5)}{3}$$

Educational objective:

Standing waves can be elicited within open and closed pipes. There are no even-numbered harmonics in the harmonic series of a pipe closed at one end because the closed end of a pipe is always a node and the open end is always an antinode.

Time spent:0 sec

Subject:
Physics

QID:402290

System:
Light and Sound